SOCIOL 723: Social Statistics II

Week 3, Maximum Likelihood: Theory

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Today

Why We Need More Than Least Squares

The Likelihood Function

Finding the Maximum

Information and the Variance of the MLE

Properties of the MLE

Likelihood-Based Tests

★ = essential, you must understand this ★ = for understanding, not examined

Why Likelihood

Two Ways to Justify an Estimator

Design / agnostic

- Start with the CEF. β is defined as the best linear approximation to it, whatever the world looks like.
- Assumes almost nothing about how Y was generated.
- This was Weeks 1–2.

Model-based

- Write down a full *probability model* for how Y was generated, with unknown parameters.
- Estimate the parameters that make the observed data most probable.
- This is maximum likelihood, and it is Weeks 3–4.

Neither is "the right one"

The design-based view is safer for causal estimands. The model-based view buys you things the agnostic view cannot deliver: predictions on the right support, structural parameters, latent variables, and outcomes that are not real numbers. Be pragmatic, and be explicit about which you are using.

OLS Is Already a Maximum Likelihood Estimator ★

Suppose we assume the full model

$$Y_i = X_i'\boldsymbol{\beta} + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2) \quad \Longrightarrow \quad Y_i \sim \mathcal{N}(X_i'\boldsymbol{\beta},\ \sigma^2).$$

Then the probability of the observed sample, as a function of the parameters, is

$$\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - X_i'\boldsymbol{\beta})^2}{2\sigma^2}\right).$$

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Taking logs, everything except $-\frac{1}{2\sigma^2}\sum_i (Y_i - X_i'\boldsymbol{\beta})^2$ is free of $\boldsymbol{\beta}$. Maximizing over $\boldsymbol{\beta}$ is exactly minimizing the sum of squared residuals.

So OLS under normality *is* the MLE. What we do today is take the same idea and apply it when the outcome is not normal.

Why We Need the Generalization

The normal linear model assumes $Y_i \in \mathbb{R}$ with constant variance. Sociological outcomes are often not like that.

- **Binary**: employed or not, voted or not, married or not. A linear model can predict probabilities below 0 and above 1.
- **Counts**: number of children, arrests, publications. Non-negative integers, right-skewed, variance tied to the mean.
- **Categorical**: occupation, party, religious affiliation. There is no meaningful numeric coding of {professional, service, manual}.
- **Ordered**: strongly agree ... strongly disagree. The gaps between categories are not known.
- **Durations, censored, truncated**: time to first birth; income top-coded at $200,000.

Maximum likelihood handles all of these, and is the engine behind almost every model in the sociological literature.

Where We Are Going

1. **The likelihood**: what it is, and the crucial distinction between probability and likelihood.
2. **Finding the maximum**: the score function, the first-order condition, and worked examples you can do on paper.
3. **Information**: how the *curvature* of the log-likelihood tells you how precise your estimate is.
4. **Properties**: consistency, asymptotic normality, efficiency, invariance, and what can go wrong.
5. **Testing**: the likelihood ratio, Wald, and score tests, which are three ways of measuring the same distance.

As in Weeks 1–2, everything examinable is done in the one-parameter scalar case. Frames marked ★ are the essential core; frames marked ★ generalize.

The Likelihood

Probability Versus Likelihood ★

Both start from the same object, the density (or mass function) $f(y \mid \theta)$. What differs is *what you hold fixed.*

Probability

$$f(y \mid \theta)$$

θ is *fixed and known*; y varies.

"If the coin is fair, how likely am I to see 7 heads in 10 tosses?"

Likelihood

$$\mathcal{L}(\theta; y) \equiv f(y \mid \theta)$$

y is *fixed* (it is your data); θ varies.

"Given that I saw 7 heads in 10 tosses, which values of θ make that outcome most probable?"

The single most important sentence today

The likelihood is *the same formula read backwards*: a function of the parameter, with the data nailed down. It is not a probability distribution over θ; it does not integrate to one, and $\mathcal{L}(\theta)$ is not "the probability that θ is true."

A First Example: Marbles in a Bag

An opaque bag contains blue and white marbles. You want the proportion θ that are blue. You draw 10 marbles with replacement and get 7 blue.

The number of blue draws is $\text{Binomial}(10, \theta)$, so

$$\mathcal{L}(\theta) = \Pr(X = 7 \mid \theta) = \binom{10}{7}\theta^7(1-\theta)^3.$$

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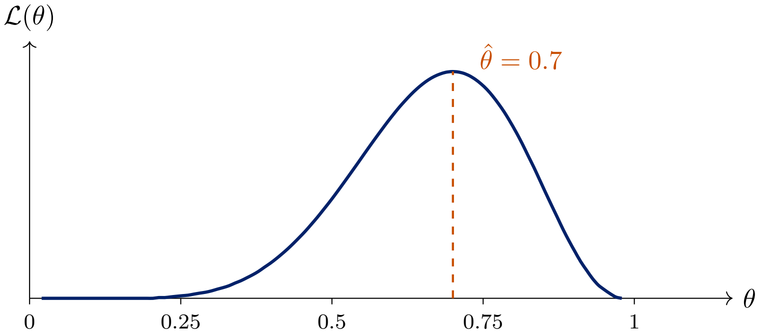
- Try $\theta = 0.3$: $\mathcal{L} = 120(0.3)^7(0.7)^3 \approx 0.0090$.
- Try $\theta = 0.5$: $\mathcal{L} = 120(0.5)^{10} \approx 0.117$.
- Try $\theta = 0.7$: $\mathcal{L} = 120(0.7)^7(0.3)^3 \approx 0.267$.

Among these, $\theta = 0.7$ makes the data we actually saw most probable. The MLE is the value that does best among *all* $\theta \in [0, 1]$.

The Likelihood, Plotted



- The height at any θ is the probability of *these* data if that θ were true.
- The peak is the MLE. The width of the peak will turn out to be our measure of uncertainty; that is the whole of Section 4.

From One Observation to Many ★

With n observations we need the *joint* density $f(y_1, \ldots, y_n \mid \theta)$. If the observations are independent given θ, it factors:

$$\mathcal{L}(\theta; y_1, \ldots, y_n) = \prod_{i=1}^{n} f(y_i \mid \theta).$$

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Products of many small numbers underflow and are painful to differentiate, so we take logs:

$$\boxed{\ell(\theta) \equiv \log \mathcal{L}(\theta) = \sum_{i=1}^{n} \log f(y_i \mid \theta)}$$

- log is strictly increasing, so it does not move the maximum.
- A product becomes a sum, so the derivative is a sum, which is why every MLE formula looks like a sample average.

Worked Example 1: Bernoulli ★

Let $Y_i \in \{0, 1\}$ with $\Pr(Y_i = 1) = \theta$. The mass function written as a single formula:

$$f(y_i \mid \theta) = \theta^{y_i}(1-\theta)^{1-y_i}.$$

(Check: at $y_i = 1$ it gives θ; at $y_i = 0$ it gives $1-\theta$.)

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum_i y_i}(1-\theta)^{n-\sum_i y_i},$$

$$\ell(\theta) = \Big(\sum_i y_i\Big)\log\theta + \Big(n - \sum_i y_i\Big)\log(1-\theta).$$

Notice that the data enter only through $\sum_i y_i$: it is a *sufficient statistic*.

Worked Example 2: Poisson

Let Y_i be a count with $\Pr(Y_i = y \mid \theta) = \dfrac{\theta^y e^{-\theta}}{y!}$.

$$\ell(\theta) = \sum_{i=1}^{n} \log\left(\frac{\theta^{y_i} e^{-\theta}}{y_i!}\right) = \sum_{i=1}^{n} \Big(y_i \log\theta - \theta - \log(y_i!)\Big)$$

$$= \log(\theta) \sum_i y_i \; - \; n\theta \; - \; \sum_i \log(y_i!).$$

- The last term does not involve θ at all. It is an *additive constant* and can be dropped for maximization, but not if you want to compare log-likelihood *values* across models.
- Again the data enter only through $\sum_i y_i$.

Worked Example 3: Normal

Let $Y_i \sim \mathcal{N}(\mu, \sigma^2)$, two unknown parameters.

$$f(y_i \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mu)^2}{2\sigma^2}\right),$$

so

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2.$$

- Look at the last term: for fixed σ^2, maximizing over μ means minimizing $\sum_i (y_i - \mu)^2$. Least squares falls out of the normal likelihood, not the other way round.
- Replacing μ with $X_i'\beta$ gives the linear regression model.

Finding the MLE

The Score Function ★

Definition

The score is the derivative of the log-likelihood with respect to the parameter:

$$S(\theta) \equiv \frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial \log f(y_i \mid \theta)}{\partial \theta}.$$

- The MLE $\hat{\theta}$ solves $S(\hat{\theta}) = 0$, the first-order condition, exactly as the normal equations were the FOC for least squares.
- Second-order condition: $\partial^2 \ell / \partial \theta^2 < 0$ at $\hat{\theta}$, so that we have a maximum rather than a minimum.
- A key population fact: $E[S(\theta_0)] = 0$ at the true parameter. The score is a *mean-zero* random variable, which is why the CLT will apply to it.

Bernoulli: Deriving $\hat{\theta}$ ★

$$\ell(\theta) = \left(\sum_i y_i\right) \log\theta + \left(n - \sum_i y_i\right) \log(1-\theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta} \overset{!}{=} 0.$$

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Multiply through by $\theta(1-\theta)$:

$$(1-\theta)\sum_i y_i - \theta\left(n - \sum_i y_i\right) = 0$$

$$\sum_i y_i - \theta\sum_i y_i - n\theta + \theta\sum_i y_i = 0$$

$$\boxed{\hat{\theta} = \frac{1}{n}\sum_i y_i = \bar{y}}$$

The MLE of a probability is the sample proportion, derived, not assumed.

Poisson: Deriving $\hat{\theta}$

$$\ell(\theta) = \log(\theta) \sum_i y_i - n\theta - \sum_i \log(y_i!)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - n \overset{!}{=} 0 \qquad \Longrightarrow \qquad \boxed{\hat{\theta} = \bar{y}}$$

Check the second-order condition:

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} < 0,$$

so we really have a maximum.

Both examples give $\bar{y}$, which is not a coincidence: for these distributions the mean *is* the parameter.

Normal: Deriving $\hat{\mu}$ and $\hat{\sigma}^2$

Two parameters, so two first-order conditions.

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (y_i - \mu) \stackrel{!}{=} 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{y}.$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (y_i - \mu)^2 \stackrel{!}{=} 0 \quad \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (y_i - \hat{\mu})^2.$$

Normal: Deriving $\hat{\mu}$ and $\hat{\sigma}^2$

Two parameters, so two first-order conditions.

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The MLE of σ^2 is biased

It divides by n, not $n-1$. So $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2 < \sigma^2$.

This is our first warning: maximum likelihood does not promise unbiasedness. What it promises is good behaviour as n grows.

When There Is No Closed Form

Most likelihoods you will meet, logit, probit, negative binomial, anything with covariates, have no algebraic solution to $S(\theta) = 0$. We search numerically.

Newton–Raphson

Start at $\theta^{(0)}$ and iterate

$$\theta^{(t+1)} = \theta^{(t)} - \left[\left. \frac{\partial^2 \ell}{\partial \theta^2} \right|_{\theta^{(t)}} \right]^{-1} S(\theta^{(t)}),$$

i.e. take a step whose size is the score divided by the curvature. Steep curvature $\Rightarrow$ small confident steps; flat $\Rightarrow$ large steps.

Variants you will meet: **BFGS** (approximates the second derivative), **Fisher scoring** (uses the expected rather than observed second derivative; this is what `glm()` does, as iteratively reweighted least squares).

Practical Matters in `R`

- `optim()` minimizes. MLE is a maximization problem, so you pass it the *negative* log-likelihood.
- Supply `par`, the starting values. Poor starting values are the most common cause of failed convergence.
- Always check the convergence code (`$convergence == 0`), a returned number is not the same as a converged number.
- Ask for `hessian = TRUE`; you need it for standard errors. Remember the sign: if you minimized $-\ell$, then the returned Hessian is $-\partial^2\ell/\partial\theta\partial\theta'$, which is already the observed information.
- Try several starting values. If they land in different places, your likelihood is multimodal or your model is not identified.

All of this is Thursday's lab.

Identification

Definition

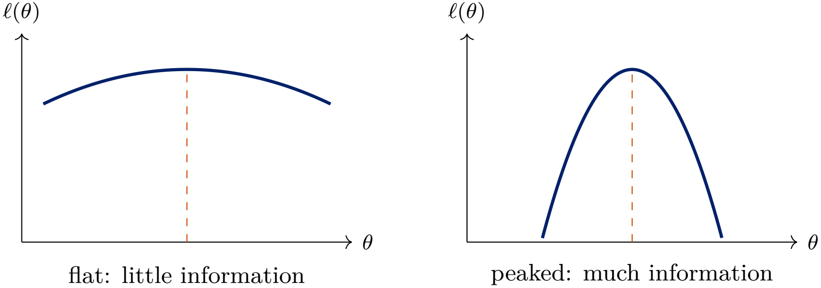
θ is identified if distinct parameter values imply distinct distributions of the data: $\theta^* \neq \theta \Rightarrow f(\cdot \mid \theta^*) \neq f(\cdot \mid \theta)$.

Non-identification means the likelihood is *flat* along some direction: the data cannot distinguish among a set of parameter values.

- **Classic example**: $Y_i \sim \mathcal{N}(\theta_1 - \theta_2, 1)$. Only the *difference* is identified. Optimizers started at different points will return different $(\hat{\theta}_1, \hat{\theta}_2)$ lying along a ridge, all with the same log-likelihood.
- This is the likelihood analogue of perfect collinearity in OLS, and it has the same symptom: a singular Hessian, so no standard errors.
- Diagnosis: multiple start values, and inspecting the Hessian's rank.

Information

Curvature Is Information ★



Both log-likelihoods peak at the same $\hat{\theta}$. But on the left, many nearby values of θ fit almost as well; we are uncertain. On the right, moving away from $\hat{\theta}$ costs a lot of fit; we are confident.

The second derivative measures exactly this.

Fisher Information ★

Definition (one parameter)

$$\mathcal{I}(\theta) \equiv -E\left[\frac{\partial^2 \ell(\theta)}{\partial \theta^2}\right] = E\left[\left(\frac{\partial \ell(\theta)}{\partial \theta}\right)^2\right] = \text{Var}\left(S(\theta)\right).$$

The second equality is the information equality, it holds when the model is correctly specified, and it will matter a great deal in a moment.

- The minus sign is there because the log-likelihood is concave at a maximum, so the second derivative is negative.
- Information is *additive over independent observations*: with n i.i.d. observations, $\mathcal{I}_n(\theta) = n\,\mathcal{I}_1(\theta)$. This is the source of the familiar $1/n$ in variances.
- Two estimators are available: the *expected* information $\mathcal{I}(\hat{\theta})$ and the *observed* information $-\partial^2\ell/\partial\theta^2$ evaluated at $\hat{\theta}$. They are asymptotically equivalent under correct specification, and software typically reports the observed version.

The Variance of the MLE ★

$$\boxed{\hat{V}(\hat{\theta}) = \mathcal{I}(\hat{\theta})^{-1} \qquad \text{or} \qquad \left[-\frac{\partial^2 \ell}{\partial \theta^2}\bigg|_{\hat{\theta}} \right]^{-1}}$$

Read it as a sentence: *the variance of the estimate is the reciprocal of the curvature of the log-likelihood at its peak.*

These are two *different* estimators, expected and observed information, equal only asymptotically and under correct specification. Pick one and say which.

- Sharper peak ⇒ larger information ⇒ smaller variance.
- With several parameters ★, $\mathcal{I}$ is a matrix and $\hat{V}(\hat{\boldsymbol{\theta}}) = \mathcal{I}(\hat{\boldsymbol{\theta}})^{-1}$; the standard errors are the square roots of its diagonal.
- This is why `optim(..., hessian = TRUE)` is not optional.

Worked: Bernoulli Information

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta},$$

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} - \frac{n - \sum_i y_i}{(1-\theta)^2}.$$

Taking expectations, using $E[\sum_i y_i] = n\theta$:

$$\mathcal{I}(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta} + \frac{n}{1-\theta} = \frac{n}{\theta(1-\theta)}.$$

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$$\mathrm{Var}(\hat{\theta}) = \mathcal{I}(\theta)^{-1} = \frac{\theta(1-\theta)}{n}. \quad \checkmark$$

Exactly the variance of a sample proportion you already know, now *derived* from the shape of the likelihood.

Worked: Poisson Information

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} \quad \Longrightarrow \quad \mathcal{I}(\theta) = \frac{E[\sum_i y_i]}{\theta^2} = \frac{n\theta}{\theta^2} = \frac{n}{\theta},$$

so

$$\mathrm{Var}(\hat{\theta}) = \frac{\theta}{n}, \qquad \hat{V}(\hat{\theta}) = \frac{\bar{y}}{n}.$$

Sanity check by direct calculation: $\hat{\theta} = \bar{y}$ and $\mathrm{Var}(\bar{Y}) = \mathrm{Var}(Y)/n = \theta/n$ because the Poisson mean equals its variance. ✓

Two routes, same answer. When you can compute the variance directly, do so; it is a free check on your information calculation.

Properties

The Four Properties ★

Under regularity conditions, as $n \to \infty$:

1. **Consistency**: $\hat{\theta} \xrightarrow{p} \theta_0$.
2. **Asymptotic normality**: $\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta_0, \mathcal{I}(\theta_0)^{-1})$.
3. **Asymptotic efficiency**: among *regular* estimators, and when the model is correctly specified, none has a smaller asymptotic variance , the MLE attains the Cramér–Rao lower bound. This is much stronger than Gauss–Markov, which only ranked *linear* unbiased estimators.
4. **Invariance**: if $\hat{\theta}$ is the MLE of θ, then $c(\hat{\theta})$ is the MLE of $c(\theta)$ for any continuous $c(\cdot)$. Wonderfully convenient: the MLE of the odds is the odds of the MLE.

The first three are asymptotic, none of them says anything about $n = 40$. Invariance, by contrast, holds exactly in any sample.

Why Consistency and Normality Hold ★

The argument is a first-order Taylor expansion of the score around θ_0:

$$0 = S(\hat\theta) \approx S(\theta_0) + \frac{\partial S(\theta_0)}{\partial \theta}(\hat\theta - \theta_0) \implies \sqrt{n}(\hat\theta - \theta_0) \approx \frac{\frac{1}{\sqrt{n}} S(\theta_0)}{-\frac{1}{n}\partial S(\theta_0)/\partial\theta}.$$

- The **denominator** is a sample average of second derivatives; by the LLN it converges to $\mathcal{I}_1(\theta_0)$.
- The **numerator** is a normalized sum of the i.i.d. mean-zero scores; by the CLT it is asymptotically $\mathcal{N}(0, \mathcal{I}_1(\theta_0))$.
- Combining: $\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta_0)^{-1})$.

This is structurally identical to the OLS argument in Week 2: a sample average in the numerator, a sample average in the denominator, LLN below and CLT above.

Regularity Conditions, and When They Fail

The theory needs: a correctly specified and identified model; a parameter space in which θ_0 is an *interior* point; a support for y that does not depend on θ; and enough smoothness to differentiate twice under the integral.

Failures you will actually meet

- **Boundary**: testing whether a variance component is zero. The usual χ^2 reference distribution is wrong.
- **Support depends on θ**: $Y_i \sim U[0, \theta]$ gives $\hat{\theta} = \max_i Y_i$, which is *always* below θ, badly biased and not asymptotically normal.
- **Separation** in logit: the MLE runs off to infinity. Week 4.
- **Too many parameters relative to n**: incidental parameters.

Maximum Likelihood Is Not Unbiased ★

- We already saw it: $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_i (y_i - \bar{y})^2$ has $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$.
- $Y_i \sim U[0, \theta]$: the MLE is $\max_i Y_i$, which is below θ with probability one.
- And unbiasedness does not survive transformation: even if $\hat{\theta}$ is unbiased for θ, $c(\hat{\theta})$ is generally biased for $c(\theta)$. (Invariance holds for the *MLE property*, not for unbiasedness.)

The trade we are making

Give up exact unbiasedness in finite samples; get consistency, asymptotic efficiency, and a completely general recipe that works for any probability model you can write down.

When the Model Is Wrong: the Sandwich Returns ★

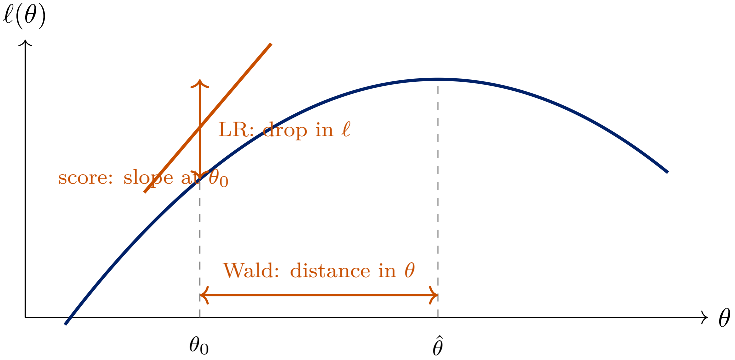
The information equality $-E[\partial^2\ell/\partial\theta^2] = E[S(\theta)^2]$ holds only if the model is correctly specified. When it is not, the two sides differ and

$$\mathrm{Var}(\hat{\theta}) \approx \underbrace{\big[-E\partial^2\ell\big]^{-1}}_{\text{bread}}\underbrace{E\big[S(\theta)S(\theta)'\big]}_{\text{meat}}\underbrace{\big[-E\partial^2\ell\big]^{-1}}_{\text{bread}}.$$

- Exactly the sandwich from Week 2, with the score in place of $X_i\epsilon_i$. It is called the *quasi-* or *pseudo-MLE* variance, or the Huber–White variance for likelihood models.
- In `R`: `sandwich::vcovHC()` works on `glm` objects too, and `vcovCL()` clusters them.
- The estimator $\hat{\theta}$ then converges to the parameter of the *best-approximating* model in the Kullback–Leibler sense, the likelihood analogue of “best linear approximation to the CEF.”

Testing

Three Ways to Measure the Same Distance



All three ask: *is* θ_0 *far from* $\hat{\theta}$*?* They are asymptotically equivalent but can differ in finite samples.

The Likelihood Ratio Test ★

Compare the maximized log-likelihood with and without the restriction:

$$LR = -2\left[\ell(\hat{\theta}_{\text{restricted}}) - \ell(\hat{\theta}_{\text{unrestricted}})\right] \xrightarrow{d} \chi^2_q,$$

where q is the number of restrictions.

- Logic: if the restriction is true, imposing it should not cost much fit.
- The restricted likelihood is always weakly *lower*, so $LR \geq 0$.
- **Requires estimating both models.** Usually easy, and generally the best-behaved of the three in finite samples, prefer it when you can.
- R: `lmtest::lrtest(m0, m1)`, or `anova(m0, m1, test = "Chisq")` for `glm`.
- The models must be nested and fitted on the same observations. Silently dropping different rows to missingness invalidates the test.

The Wald Test ★

Measure the horizontal distance, scaled by the estimated standard error:

$$W = \frac{(\hat{\theta} - \theta_0)^2}{\hat{V}(\hat{\theta})} \xrightarrow{d} \chi^2_1, \qquad \text{or} \quad z = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}.$$

- This is the t-ratio you already read off every regression table.
- **Requires only the unrestricted model**, convenient when the restricted model is hard to fit.
- Multi-parameter version ★: $W = (\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})'[\boldsymbol{R}\hat{V}(\hat{\boldsymbol{\theta}})\boldsymbol{R}']^{-1}(\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})$, identical in form to Week 2.
- Caution: the Wald test is not invariant to how you parameterize the restriction. Testing $\theta = 1$ and $\log\theta = 0$ can give different answers. The LR test does not have this problem.

The Score (Lagrange Multiplier) Test

Measure the slope of the log-likelihood *at the null*:

$$LM = \frac{S(\theta_0)^2}{\mathcal{I}(\theta_0)} \xrightarrow{d} \chi^2_q.$$

- Logic: if θ_0 were the maximum, the slope there would be zero. A steep slope is evidence against it.
- **Requires only the restricted model**, useful when the unrestricted model is expensive to fit, e.g. testing for overdispersion without fitting the negative binomial.
- Many familiar specification tests are score tests in disguise, including Breusch–Pagan for heteroskedasticity.
- **Which to use?** Asymptotically the three are identical. In finite samples: LR if you can fit both models, Wald if only the unrestricted is available, score if only the restricted is.

Model Comparison Beyond Nested Tests ★

LR, Wald and score all require *nested* models. For non-nested comparison:

$$AIC = -2\ell(\hat{\theta}) + 2k, \qquad BIC = -2\ell(\hat{\theta}) + k \log n.$$

- Both penalize fit by complexity; lower is better.
- BIC penalizes more heavily once $n > 7$, and increasingly so with n. BIC targets the "true" model if it is in the candidate set; AIC targets predictive accuracy.
- Neither is a hypothesis test; there is no p-value and no threshold.
- Comparable only across models fitted on *exactly the same* data with the same outcome scale.

We return to model selection properly in Week 12, where cross-validation replaces these analytic penalties.

Where This Leaves Us

- A likelihood is the data's probability read as a function of the parameter. The MLE maximizes it; the score is the derivative you set to zero.
- The *curvature* of the log-likelihood is information, and its inverse is the variance. Everything about precision follows from the shape of the peak.
- The MLE is consistent, asymptotically normal, and efficient, but not unbiased, and all of the guarantees are asymptotic.
- LR, Wald, and score are three views of the same question.

Next

Thursday: code a likelihood and maximize it numerically in `R`.
Week 4: put covariates inside the likelihood, the generalized linear model, and the limited dependent variable models you will actually use.

References

- Pawitan, Yudi. 2013. *In All Likelihood: Statistical Modeling and Inference Using Likelihood.* Oxford University Press. [Ch. 2 and 4 required; Ch. 3 additional]
- Casella, George and Roger L. Berger. 2002. *Statistical Inference*, 2nd ed. Duxbury. [Ch. 7 and 10, for a fuller treatment]
- King, Gary. 1998. *Unifying Political Methodology: The Likelihood Theory of Statistical Inference*. University of Michigan Press.
- White, Halbert. 1982. "Maximum Likelihood Estimation of Misspecified Models." *Econometrica* 50(1): 1–25. [the quasi-MLE sandwich]